PREFACE

This role challenge was designed to be comprehensive, and going through it has forced me to think deeply about clinical workflows, operational constraints, tech systems, and economic trade-offs in healthcare. It was hard but absolutely worth it, and it really has opened my eyes to how actual healthcare actually gets practiced in the real world, rather than just at a high level.

In going through this role challenge, I've employed a combination of structured research, publicly available literature, domain-specific articles, product reasoning, and iterative synthesis. I've included hyperlinks wherever possible to reference external sources. Where there wasn't direct data or external references, I've relied on logical reasoning, first principles, and educated hypotheses based on patterns I've observed in healthcare systems.

I want to emphasize that this exercise is not just about "solving" the questions but really understanding the underlying problems, constraints, and trade-offs that clinicians, operators, and healthcare systems face.

Finally, I'd like to thank the team that created this role challenge. The level of detail in the questions has forced me to think deeply and honestly about these topics, and it's been one of the most valuable interview experiences I've had.

Practice Type Overview

Deliverable A: Practice Type Taxonomy Table

Parameter	IDNs	Private Practice
Typical ownership and organisational model **Source: IDNs, Private practice**	• Consortium(A group of independent organizations jointly owning or governing a shared healthcare entity) • Parent holding(A central holding company owns multiple healthcare entities but allows partial operational independence) • Corporate model(A single corporate organization fully owns and centrally controls all operations and assets) • Most of it is centralized(especially System IV IDNs)	• Broadly classified into **sole proprietorship**, **partnership**(General,limited,limited liability), **LLC** and **PLLC**, **Corporations**(C-corp, S-corp,non-profit)
Roles responsible for performing the ultrasound scan	• **Diagnostic medical sonographers** in general, they are more preferred if they have a ARDMS(American Registry for Diagnostic Medical Sonography) certification or if they went through a CAAHEP(The Commission on Accreditation of Allied Health Education Programs) accredited training program • **Maternal-Fetal Medicine (MFM) Sonographer** • **Physicians**(some select cases) • Sometimes **Physician assistants** (PAs), **Nurse Practitioners** (NPs), and **Certified Nurse-Midwives**	• Can be anyone described for the IDN column

	(CNMs) may perform point of care ultrasound or limited ultrasound	
Roles responsible for interpreting and signing off on the scan	• Primarily Radiologists • OB-GYN physicians • MFM physicians	• Can vary based on the hiring power of the private practice • Typically this would be the same as that in IDNs or anyone with a valid medical license. • It can also be teleradiologists/service providers.
Staffing, operational, and workflow constraints **Source: IDNs, Private Practices**	From the references that I have gathered and mentioned below some of the constraints that are generally seen in IDNs are as follows. (NOTE: This is not particularly catered to pre-natal care but can apply to it) • According to this article, high workload and stress lead to burnout, increasing error rates, reducing productivity, and causing frequent staff turnover that disrupts care continuity and raises training costs. • Doctors and staff resist new standardized workflows, Physicians are used to working independently, so adapting to common rules and shared workflows across multiple hospitals is difficult and causes friction. • Too many systems that don't talk to each other Different hospitals and clinics inside an IDN use different IT systems, making it hard to share patient	Private practices also have the same constraints that is seen in IDNs but some of the notable constraints that differ from IDNs are: • They generally have small teams,when one staff member quits or is absent, billing, scheduling, and patient care are immediately disrupted. • According to this article , staff do multiple work so the same person may handle reception, insurance verification, and billing. This leads to burnout and more mistakes, whereas IDNs can assign specialized roles. • Each staff member may follow their own way of doing intake, documentation, and follow-ups. IDNs usually enforce standardized protocols across sites resulting in inconsistent workflow

	information smoothly and slowing down care delivery. ● Complex coordination across many locations IDNs are large and spread across regions, so managing care between hospitals, clinics, labs, and home care services becomes operationally complicated. ● Credentialing and compliance take a lot of time Every staff member must meet licensing, background checks, and regulatory rules, which slows down onboarding and creates operational delays. ● Leadership and decision-making move slowly Because IDNs are big organizations, approvals for changes (new tools, new workflows, new vendors) take time and involve many stakeholders.	● Private practices often use outdated or disconnected software for records and billing, while IDNs invest in enterprise-level EHR and integrated systems. ● Constant updates in insurance rules and medical codes overwhelm small admin teams, whereas IDNs have compliance and legal departments to manage this which makes billing,coding and regulatory changes hard to keep up with ● According to this article, there is poor cash flow and revenue in US private OB/GYN practices ● In private practices, change depends heavily on the doctor's time and willingness hence there is a strong dependence on one or two physicians for decisions.
Constraints related to technology adoption and integration **Source: IDNs, Private Practices**	Although the U.S. health system is large and they can easily adopt or integrate new software even when it clearly improves workflow, they are not doing it because of the following constraints: ● They have an Enterprise system lock-in (EHR / PACS / RIS), which means all of the systems they use are legacy systems and anything new must be **integrated cleanly** and shouldn't disrupt	Comparatively private practices also face some of the same constraints. In my opinion, IDNs struggle because they are too big and complex. Private practices struggle because they are too small and resource-limited. The following are the constraints faced by private practices: ● Private practices cannot afford expensive enterprise software, long

	the workflow. Most of them use vendor controlled EHRs like Cerner,Epic or Meditech so API integrations would be hard which only adds to their switching cost ● The above point directly leads to integration complexity since there will be downtime while integrating new tech to an already existing system. ● Any new technology will have to follow data standards like HL7 or FHIR for sending messages between systems. DICOM image standard for storing ultrasound images in Pixels. All of these take time and hence will cause an Interoperability issue. ● There is also a Regulatory compliance that needs to be followed (HIPPA security review) that evaluates technical safeguards, Administrative safeguards and physical safeguards. These compliances make it harder to integrate new technology ● There is always a simple burden of training staff to adopt new technology.	contracts, or large IT projects like IDNs. Even if a tool improves workflow, cost is often the biggest blocker. ● Private practices are often not connected to HIEs(they act as bridges between EHRs) or large networks, so sharing images and reports with hospitals is slow and manual. ● HIPAA rules still apply to them as well, but private practices lack legal and compliance teams. Fear of violating privacy laws makes them cautious about adopting new technology. ● Comparatively they have EMR as opposed to EHR although the term is used interchangeably,If their EMR vendor does not support integration, they are stuck. Switching vendors is risky and disruptive to daily operations.

Main difference between both:

IDNs: struggle because of legacy systems, complex integrations, and strict compliance

Private practices: struggle because of lack of money, lack of IT staff, and fragile workflows

Topic 2: Clinical Workflow Reconstruction

Deliverable B: Workflow reconstruction

- The following workflow has been reconstructed by referring to the AIUM-ACR-ACOG-SMFM-SRU Practice Parameter for the Performance of Standard Diagnostic Obstetric Ultrasound Examinations.
- Additionally you can find the figma link here, which gives a complete overview of the workflow end-to-end.
- **PLEASE NOTE:** For assumptions / ambiguities / variations are selectively written where they might exist meaningfully

STEP 1: Patient arrival, order verification, and pre-scan intake

Clinical Relevance	Personnel involved	Tools used	Attention shifts	Time spent
• The patient arrives and is identified (name, DOB, other identifiers). • Staff confirm that there is a valid written/electronic request from an appropriately licensed provider with a medical indication (e.g., dating, anatomy survey, growth, bleeding, pain), consistent with the obstetric parameter. • Relevant history (LMP, prior scans, complications) is briefly reviewed to ensure the appropriate exam type (standard, limited, or specialized) is performed.	• Front-desk/registration staff. • Sonographer. • Referring provider (indirectly, as originator of the request). • Occasionally nurse/MA to obtain vitals.	EHR for verification and RIS system for scheduling	screen ↔ patient↔keyboard.	**10-15 min** (± 5mins)

Assumptions:

- Order is already in the EHR and meets AIUM requirement for sufficient information to allow appropriate performance and interpretation
- Time spent can sometimes vary far too long, heavily site dependent

Variations:

- After going through the sources(link mentioned in the References), In small/private practices, front-desk/reception roles commonly span registration, scheduling,

insurance/eligibility checks, payment collection, and often billing support, and sometimes even managing outbound communication of results or records to patients.

- In large IDNs, those functions are more often segmented (separate scheduling center, revenue cycle department, imaging records team).

STEP 2: Pre-scan review, consent, and safety setup

Clinical Relevance	Personnel involved	Tools used	Attention shifts	Time spent
• Sonographer reviews indication, prior imaging, and relevant clinical information (ex- previous accurate dating scan, high-risk conditions) to decide whether the requested exam is standard, limited, or specialized per obstetric parameter definitions. • Clarifies if any additional specialized components are anticipated (NT, cervical length, Doppler, BPP, or procedure guidance) • The patient is positioned (supine/semi-recumbent, left tilt later in pregnancy if needed) and abdomen exposed for transabdominal or prepared for transvaginal exam. • Sonographer selects the appropriate preset and transducer (ex- ≥3 MHz abdominal, transvaginal early in pregnancy), verifies output display standards (TI, MI) are visible, and ensures adherence to ALARA and fetal safety recommendations, including avoiding unnecessary spectral Doppler, especially in early gestation.	• Sonographer, Chaperone as per local policy • Physician to consult in case of any ambiguities	EHR (clinical notes, lab results, prior reports). Ultrasound machines (presets, TI/MI display) and Infection control supplies.	probe ↔ console control ↔montiors.	**15-20 mins** (± 5mins)

Considerable variations can be seen between both practice types in this step:

1. Equipment quality control is handled separately according to accreditation standards.
2. CDC standards for HLD(high-level-disinfection) have to be followed and may vary in both practice types.

3. Moreover, assuming that due to less staff the physician themselves, might perform this step in private practices

STEP 3: Real-time scanning and measurements

Clinical Relevance	Personnel involved	Tools used	Attention shifts	Time spent
● Sonographer performs an initial sweep: ○ First trimester: uterus, gestational sac(s), yolk sac, embryo/fetus, and uterus/cervix/adnexa/cul-de-sac are scanned. Fetal number and amnionicity is documented for multiple pregnancy. See if there is a need for specialized ultrasound to detect anatomical abnormalities ○ Later gestation: fetal number, cardiac activity, presentation, placenta location, and a qualitative feel for amniotic fluid is scanned ● Fetal cardiac activity is documented with 2D video or M-mode, not routine pulsed Doppler, as recommended. ● transvaginal cervix assessment maybe performed in later gestation if indicated or poorly visualized.	● Sonographer (with possible physician presence in complex/urgent situations).	Ultrasound machine (2D, M-mode,color Doppler).	Continuous probe ↔ screen loop, intermittent keyboard use for measurement, label, and image storage.	**15-25 mins** (± 15mins) Late trimesters will take more time

Assumptions:

1. Four way movement of the probe(sliding,rotating,dipping,angling) is assumed to be done.

Variations and Ambiguity:

1. Clinically there might be no ambiguity between both practice types because both may follow the same AIUM standards. The ambiguity lies in the fact about **how decisions are made during the sweep** and who decides during the scan that a detailed / specialized exam is needed?
2. The time mentioned varies depending on the trimester the female is in. According to sources online, time spent for an ultrasound is generally 30 minutes but if we follow strict ISUOG standard it varies widely depending on numerous reasons like the amount of Amniotic fluid, weight of the person, size of the fetus and its position. Moreover, they are always good

chances of not obtaining a good result of the image due to which the sonographer has to redo the scan again

Likely difference

- **IDNs**:Sonographer flags concern → physician reviews later → separate detailed scan ordered
- **Private practices**:Physician may be present or immediately available → decision made in real time

This is **operational**, not clinical.

3. Ambiguity lies in the threshold of switching from transabdominal to transvaginal assessment. This is on the workflow level and not the guideline level.
    a. In **IDNs**: Strict criteria and documentation is required
    b. In **private practices**: More discretion based on clinician judgment and time constraints
4. Order of structures scanned; some sonographers follow head-to-toe, others organ-system-based—all acceptable if minimum elements are covered.

STEP 4: Biometry, gestational age, and growth

Clinical Relevance	Personnel involved	Tools used	Attention shifts	Time spent
• Different types of measurements are taken to get the gestational age. Few notable measurements are: CRL(crown rump length), BPD(biparietal diameter), HC and AC(abdominal and head circumference) are taken and the fetal weight is compared against standard growth charts. • Amniotic fluid evaluation using qualitative assessment or semiquantitative methods (deepest vertical pocket or AFI), with defined thresholds for oligohydramnios and polyhydramnios are also done. • Targeted measurements such as nuchal translucency and cervical	• Sonographer	Ultrasound machine with its presets and fetal weight calculators.	probe ↔ screen (caliper placement) ↔ console for saving and labeling measurements There is more focus on angle, sample volume placement, and documentation.	**5-20mins** (± 15mins) Depends on what is added

length, performed using strict technical criteria and standardized transvaginal techniques when indicated. ● Adjunct imaging and procedures (ex- Doppler studies, biophysical profile, or ultrasound-guided procedures like amniocentesis) performed only when clinically indicated, with appropriate consent and documentation.				

Assumptions:

1. Probe sweeping is assumed to be done using ISUOGs 20 planes and 2 sweep approach

Variations and Ambiguity:

1. Ambiguity may exist for decision escalation, when we want to move from standard measurement to specialized or limited measurement
    a. In **IDNS**: escalation often requires a new order or protocol-defined trigger
    b. In **Private Practices:** escalation may happen immediately during the same visit
2. It can also lie in type of Amniotic fluid assessment method used in practice
    a. In **IDNS**: stronger preference for documented semiquantitative method
    b. In **Private Practices:** qualitative assessment may be used more often unless abnormality suspected

STEP 5: On-machine documentation and image curation

Clinical Relevance	Personnel involved	Tools used	Attention shifts	Time spent
● Verification and archiving of images ensuring all required normal and abnormal findings are captured, with each image or cine loop labeled with patient identifiers, facility details, exam date/time, output indices (TI/MI when required), anatomic labels, laterality, and orientation as appropriate. ● Completion of examination documentation such as worksheets or	● Sonographer	Ultrasound system's annotation tools and DICOM metadata. PACS for storing the image	Worksheet screen/paper ↔ keyboard/pen; minimal probe use	**5-10mins** (± 5mins)

electronic forms, including patient identifiers, date/time, and the name of the individual who performed the exam and completed the documentation. ● Recording of measurements and observations including all quantitative measurements (biometry, fluid indices, cervical length) and key qualitative findings or technical limitations affecting image quality.				

Variations:

There is workflow variation but variation related to the tools used can be seen between both the practices:

1. **Tooling used**
    a. **IDNs:** PACS + EHR-native reporting; worksheets often fully electronic
    b. **Private practices:** mix of machine-generated worksheets, PDFs, or semi-manual entry. They might use softwares that are not specific to OB-GYN but a more generic one that gets the job done.
2. **Timing of documentation**
    a. **IDNs:** documentation often finalized after the patient leaves
    b. **Private practices:** documentation may be completed immediately during or right after the scan
3. **Depth of qualitative notes**
    a. **IDNs:** more structured fields and mandated phrasing
    b. **Private practices:** more free-text narrative descriptions

STEP 6: Patient debrief and transition

Clinical Relevance	Personnel involved	Tools used	Attention shifts	Time spent
● Sonographer cleans gel, helps patient sit up, and briefly informs the patient how and when formal results will be communicated. ● In obvious urgent findings (e.g., absent cardiac activity, placenta	● Sonographer. ● Patient (± partner). ● Interpreting provider if urgent.	None beyond local communication tools.	Sonographer to patient	**5-10mins** (± 5mins)

previa with active bleeding), sonographer triggers immediate provider-to-provider communication prior to discharge, as outlined in nonroutine results reporting.				

STEP 7: Image archiving and final report creation

Clinical Relevance	Personnel involved	Tools used	Attention shifts	Time spent
• Image storage and retention with all still images and cine loops archived in a retrievable electronic format, verified for diagnostic adequacy, and retained according to clinical and legal requirements. • Physician image review and adequacy assessment where the interpreting provider evaluates images and measurements against obstetric practice parameters, assesses technical limitations, compares with prior studies, and determines the need for repeat or detailed examinations. • Generation and sign-off of the final diagnostic report, which becomes the definitive medical record and includes required demographic information, exam indication, exam type, and technical details. • Structured reporting of findings and conclusions, including description of procedures performed, measurements and observations, clear anatomic terminology, summary/impression, comparison with prior imaging, follow-up recommendations, explanations for incomplete views, and documentation of any significant inter-provider communication.	• Sonographer • Radiologist, MFM or OB	PACS/VHR, image viewing software, RIS for completion of schedule	Screen ↔ keyboard; possibly back to images during reporting.	**10-30 mins** (± 10mins)

• Final report is routed/made available to the ordering provider, who is responsible for reviewing it; the ultrasound facility ensures retrievable archiving of both images and report and ensures the requesting provider has access. • The ordering provider uses the report to counsel the patient, arrange follow-up imaging (growth scans, detailed anatomy, fetal echo, procedures), or refer for further care, in line with obstetric practice parameters.				

Variations with Assumptions:

Image review and diagnostic adequacy assessment are standardized, with variation arising mainly from infrastructure, timing of review, and discretion in escalation decisions.

The following may be the variations seen:

1. Image storage & retrieval systems
    a. In IDNs: centralized PACS with enterprise retention policies
    b. In Private practices: ultrasound machine archive or lighter-weight PACS, sometimes vendor-hosted. They generally may not use specialized software for scheduling ultrasound.
2. Timing of image review
    a. In IDNs: interpretation often happens asynchronously, sometimes hours later
    b. In Private practices: interpreting providers may review images immediately after acquisition. These interpreting providers may be outsourced or in-house.
3. Comparison with prior studies
    a. In IDNs: prior imaging readily available across sites via enterprise systems
    b. In Private practices: prior studies may be external or unavailable, limiting comparison

ADDITIONAL INFORMATION: In case of non routine results when urgent findings require immediate action, a clearly labeled preliminary report and direct, timely provider-to-provider communication are used to support clinical decisions, with all communications documented and reconciled in the signed final report. The guidelines can be found here under REPORTING OF NON ROUTINE RESULTS

Workflow Analysis

Deliverable C: Time-Motion Analysis

NOTE: Time range mentioned here and in the workflow are just an educated guess and does not reflect the real-time taken.

Abbreviations:

VA- Value adding

NVA- non value adding

RW- rework

Key Assumptions:

- Standard 18–22 week anatomy scan, single fetus, outpatient setting.
- No major anomalies detected; no Doppler/BPP/procedure add-ons.
- No long delays between steps (patient is taken straight through the process).
- Data that i have used to come to a ground conclusion:
  - NHS fetal anomaly specifications: 30-minute minimum appointment for singleton anatomy scan (link)
  - Australasian Sonographers Association survey: morphology scans most commonly 40–60 minutes of sonographer time (link)
  - NHS and RadiologyInfo patient info: scan usually takes around 20–30 minutes (link)

Based on the above, I assume a typical sonographer appointment block is roughly equal to 45 minutes (30–45 min scan + 10–15 min doc/turnover), and physician time is roughly equal to 25 minutes (review + report) → approx 70 minutes total per study episode, recognizing big variation.

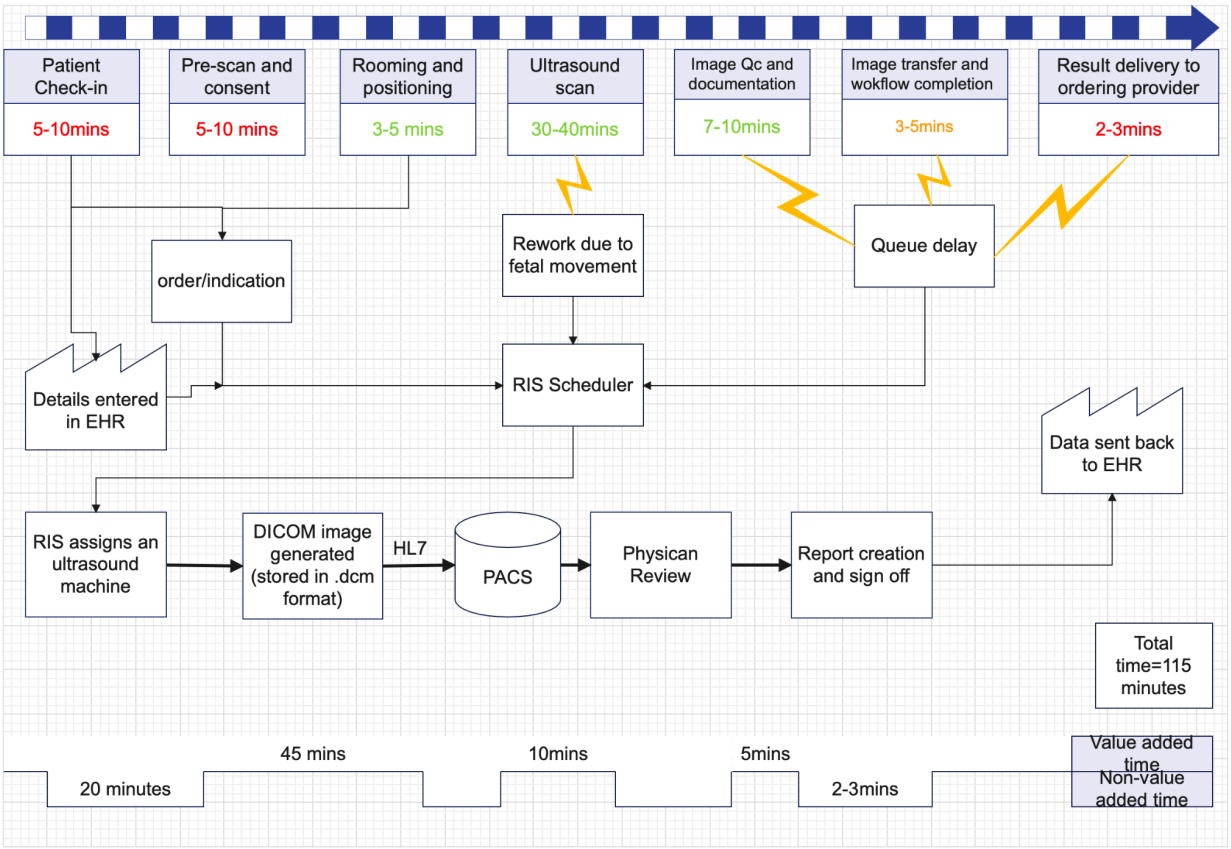
Workflow Phase	Time range(mins)	Value adding	Reasoning as to why it is VA/NVA/RW and Time assumption
Check-in, registration, ID, order confirmation	5-10	NVA	Can be made shorter with online pre registration
Pre-scan clinical review & consent (indication, brief history, explanation)	5-10	VA	The time is assumed here and it is VA because it ensures correct exam/indication and patient understanding, directly affects exam appropriateness.
Rooming, positioning, exposing abdomen	3-5	NVA	This is setup logistics, it is needed but not diagnostic. Often overlaps with Step 2.
Equipment setup & ALARA	2-4	NVA	This is already a technical set and adds

checks (probe/preset, TI/MI, Doppler off unless needed)			indirect value
Initial survey: viability, number, presentation, placenta overview, fluid impression	8-10	VA	Core diagnostic work. Time varies with BMI, fetal position.
Systematic anatomic survey	20	VA	The time assumed here is a calibrated assumption based on the data I have mentioned above. Dominant component of the scan. In easy cases closer to 15; in challenging cases 30+.
Biometry & structured measurements	8-10	VA	Measurements directly drive dating/growth decisions. Often interleaved with Step 6.
On-machine documentation & image QC (labels, identifiers, check all required views, complete worksheet)	7-10	VA	The time here is assumed and this is a VA because it ensures the exam meets AIUM documentation standards; missing views → re-scan.
Post-scan patient interaction (wipe gel, help up, explain when/how results come)	3-5	NVA	Important for experience, but does not add clinical information. This may be more prominent in private practices as i assume that they are more POCUS oriented
Image transfer to PACS, worklist completion, resolving minor IT glitches	3-5	NVA	Largely technical,sometimes automated so time spent here is almost standard throughout all practices.
Rework: repeat views / extra measurements due to inadequate images	5	RW	Only occurs in some exams; 5 min here is an average allowance (0 for easy cases, 10–15 for difficult).
Physician review of images & measurements	15	VA	Direct diagnostic decision-making. Normal straightforward cases may be closer to 10 min; complex 20–30.
Report dictation / structured report completion & sign-off	10	VA	Produces definitive documentation required by AIUM documentation parameter; time depends on templates vs free text.
Routine electronic result delivery to ordering provider (no phone call)	2-3	NVA	Electronic routing is mostly automated,minimal human time.

- Total value adding time from the above table: **58-85 minutes**

- Total non value adding time from the above table: **18-30 minutes**
- Total rework time: can be **5 minutes** but in cases of Urgent result communication in case of critical finding it can extend to **15-20 minutes**

Deliverable D: Value Stream Map



The value stream map was made using wondershare Edrawmax software. The link to its PDF is [here.]

Friction & Cognitive Load Analysis

Deliverable E: Friction Identification

ASSUMPTION: Based on the workflow that I have reconstructed, it is my understanding that there isn't much clinical difference between IDNs and private practices. Clinically both the practices have similar types of friction points, significant differences can be seen operationally.

NOTE: hyperlinks are given wherever source is mentioned

Friction Point	Underlying causes	Who Experiences it	Error,Delays and clinical risks associated with this	Operational or Cognitive
Incomplete / low-quality orders and indications **During:** Ordering provider → scheduling/EHR → sonographer (both IDN and private practice).	• According to this article Radiology request forms often contain missing or incorrect clinical information, leading to delay and rework. • Root-cause analyses of radiology turnaround time highlight errors in radiology requests and triage protocol issues as key contributors to delays.	Sonographers,Radiologists/MFMs,Ordering providers	• Delays in starting the exam while staff contact the ordering provider. • Wrong exam performed (e.g., limited vs standard vs detailed), requiring re-imaging (rework). • Misprioritization of urgent cases due to missing urgency info → delayed diagnosis	Operational
Scan time complexity and image rework **During scanning**	• According to a Large sonographer survey: 56% of sonographers doing morphology scans report insufficient time, especially for challenging cases. • I can draw an indirect conclusion from this paper which tries to capture the workflow of the process and while saying it shows the data for scan time which approximately is around 40 minutes with 11 minute variability. • According to this article Obstetric ultrasound often fails	Sonographers, Patients	• Missed or poorly characterized anomalies due to rushed or incomplete scans. • Frustration and decreased trust from patients when images are inadequate according to this article	Both

	to provide all needed information due to poor fetal position and poor ultrasonic visualization (ex- obesity), requiring repeat scans.			
Radiologist/MFM reading queue and turnaround time During: Post acquisition of image while interpreting	• According to this article Radiology studies show that operational bottlenecks, SOP adherence issues, and triage problems are major contributors to long turnaround times (TAT). • Hence, Incomplete or incorrect orders and communication breakdowns also slow TAT. • According to this article, if radiologists are outsourced and live in a different time zone, it significantly contributes to the overall TAT due to incorrect time synchronization	Radiologists/MFMs, patients	• Delayed diagnosis or intervention (e.g., placenta previa, growth restriction, fetal demise). • Increased chance that urgent findings are buried among routine cases if triage is poor. • Burnout and fatigue, which can degrade diagnostic accuracy.	Both
Manual reporting and dual data entry **During:** Sonographer worksheet → radiologist report; PACS ↔ EHR (both settings, especially where integration is incomplete).	• According to ramsoft, before full PACS–EHR integration, radiology departments relied on manual copying of data and dual entry, causing delays and discrepancies. • Such Workflow optimization articles highlight manual reporting and repetitive admin tasks as major strain and delay causes. (This is more prevalent in private practices)	Sonographers,Radiologists, clerical staff	• Transposed numbers in fetal measurements can lead to incorrect estimates of gestational age or fetal weight, affecting clinical decisions. • Incorrect labeling of laterality or placenta location may result in wrong clinical interpretations and inappropriate patient management. • Extended reporting time	Both

			creates operational delays and slows down downstream care decisions.	
Single-resource bottlenecks in private practices (one OB or one sonographer) **During:** Scanning	• Practice often has a single sonographer or OB performing all scans; any over-run cascades to subsequent patients. • External readers (cloud radiology) add a separate reading queue, which can help or hurt depending on capacity.	Sonographers, patients	• Rushed scans or less thorough reviews late in the day. • Delays in urgent add-on scans (e.g., new bleeding) if the only scanner is occupied.	Cognitive

BONUS POINT:

Sonographers have to constantly shift their attention from probe to screen and back to probe. This exposes them to awkward postures, static muscle contractions, and repetitive, precise movements of the upper limbs, especially wrists and hands.

Here is the link for a source supporting this assumption. This can be more of a physical/ cognitive friction point but still often overlooked by many.

Friction & Cognitive Load Analysis

Deliverable F: Friction Prioritisation

NOTE: The Moscow model is used for prioritisation of frictions with a slight modification. The Must Have, Should Have, Could have and Not Important just rank the frictions based on their importance(Must have- top friction, not important- low friction) but are still an existing friction problem nevertheless

MODIFIED MOSCOW MODEL FOR FRICTION PRIORITIZATION

MUST HAVE

Scan time complexity and image rework

- Directly tied to image quality, completeness, and standardization
- Its both frequent and severe
- It is realistically addressable because it fits in the workflow and does not hinder any existing regulation

SHOULD HAVE

Radiologist/MFM reading queue and turnaround time

- Turnaround time (TAT) is a key radiology KPI ,delays in this often stem from reading queues, triage issues, and communication breakdowns.
- Its frequent but severity is moderate to high
- It is realistically addressable because it fits in the workflow and does not hinder any existing regulation

COULD HAVE

Manual reporting and dual data entry

- It touches almost every exam,It directly affects data integrity.It consumes scarce expert time.
- Frequency is high but severity is moderate with spikes
- It is realistically addressable because most modern ultrasound machines, PACS, and reporting systems already support automatic transfer of measurements and key metadata into the report template.

NOT IMPORTANT

Single-resource bottlenecks in private practices

- Not our top priority but still persists as a major problem
- Frequent in private practices, severity is moderate with spikes
- Can be addressed realistically with existing workflow and regulations since this is more of an operational constraint

Practice Type Selection & GTM Rationale

Deliverable G: Practice Type Selection

IMPORTANT: Choosing any of the practice types requires more detail about the current revenue origin medical is making. **My assumption**: If you are pre-revenue then choose IDNs, if considerable revenue is already made that can sustain the business for another 1-2 years then go for private practices. Here I assume that origin medical is pre revenue and hence chose IDNs to target.

1.)Why did you select this particular practice type as the initial target?

- IDNs have more steps and roles in the imaging chain, They follow a standard SOP with clear division of labour.
- Their friction points are more defined and addressable in general. A tool like OMEA that improves scan completeness, standardization, and preliminary reporting plugs into multiple friction points at once, which is much more visible in a complex IDN workflow than in a small, tightly coupled private practice.
- It is my understanding that, the more complex the workflow, the more opportunity to remove waste and rework—and the more clearly you can demonstrate value.

2.)Why did you choose not to prioritise the other practice types at this stage?

- Largely because of how scaling of origin medical would happen if we were to target private practices. For an initial scale, we need revenue and private practices are less likely to have resources to invest – this claim is backed by this paper.
- Moreover they have Smaller scale per installation A single private practice might have 1–2 machines and a few clinicians; an IDN has dozens of machines and many providers, hence the absolute time and cost savings from reducing rework and documentation are much larger in a system doing thousands of scans per year.

3.)Where does the most significant workflow inefficiency or economic gap exist?

Scan quality and rework seems to be the most significant inefficiency but OMEA solves it with real time monitoring and exam completeness. OMEA solves the "imaging" part of the workflow (scans,read,report), but what happens after that, i.e AFTER the report hits the EHR. The time for the OB to see the report and refer to a specialist is still a grey area. But it is understandable, because that may seem out of scope to solve. In which case, I would say focusing on the ergonomic or cognitive pain point of the sonographers would be apt.

4.)Where is the return on investment most immediately and defensibly realised?

- Reduced re-scan / recall rate
- Lower reading time per case and improved TAT

5.)What key factors and considerations influenced your decision-making process?

- IDNs have a thorough and established workflow which gives us more visibility to tackle the friction points
- High-impact frictions (scan rework, TAT, dual entry) are more consequential in IDNs because they have higher volumes and clear division of labour

Practice Type Selection & GTM Rationale

Deliverable H: Go-To-Market & Regulatory Context

1.)Who is the economic buyer, and who is the primary end user of the product?

The economic buyer would be health systems in general and the primary end user of the product would be a Sonographer

2.)What core problem or value proposition would you lead with in the go-to-market narrative?

The core problem that I would lead would be to reduce prenatal ultrasound recalls and reading time leading to standardization of scans, scan completeness, exam completeness without adding staff or changing workflows.

3.)What evidence or proof points would this customer require to support adoption?

This can vary from small levels of proofs like video proof of testimonials by sonographers to big levels of proofs like clinical trials success. But the following are some notable proof points that can support adoption:

- Live integration demo (With EHR)
- FDA clearance documentation
- Pilot studies
- Any qualitative proof that convinces the IDN's board of members that the product will increase revenue

4.)At a high level, how does FDA classification influence the go-to-market strategy?

This paper Mooghali M, et al. "Medical device risk (re)classification: lessons from the FDA's 515 Program Initiative." BMJ Innovations. 2023 talks about the three levels of risk systems and their direct impact on regulatory requirements and market authorization pathways. They are as follows:

1. Class I: Fast track/low risk/general controls, No premarket review (exempt), Sell immediately after registration
2. Class II: moderate risk,510(k) clearance required, OMEA probably fits here(i'm not sure about this)
3. Class III: high risk/PMA(pre market approval)

So based on the above data this can impact how fast the product can reach the market and how long will it take to make actual sales

Trade-offs & Product Decisioning

Deliverable I: Forced Trade-offs

Primary Goal: Reduce total sonographer scanning time per exam by 20% while maintaining 100% anatomy completeness compliance

1.)Why did I prioritise this objective over the alternatives?

I have identified that Sonographer time is the bottleneck. IDNs are staffing-constrained – they cannot hire more sonographers quickly (training takes 2+ years). If we can reduce overall scan

time then we can focus more on patient care and improve patient experience (something that is seen better in Private practice)

2.)What did I intentionally deprioritize?

Radiologist/MFM reading time was intentionally deprioritized as a result of prioritising the above friction point because, although TAT is a key KPI, this part of the workflow is directly influenced by the primary friction point that I have mentioned. HOW? Sonographer time = capacity constraint and radiologist time = efficiency constraint, if we automatically reduce Sonographer's time scan; this friction point can be addressed downstream as a result of the former's cascading effect.

3.)What second-order risks or unintended consequences could arise from this decision?

If we use AI to reduce scan time, it might increase the reliance of the sonographer on the product which can hinder clinical judgement. AI can make mistakes and hence if its accuracy is wrong in some cases then it will affect patient care plan downstream

Solution Framing (System-Level)

Deliverable J: Solution as a System

Friction: Inadequate scan time and image rework

Solution overview:

Since OMEA already seems to solve this issue ensuring scan completeness, we can probably add features that can enhance sonographer's experience by enabling ISUOG biometry and 20 + 2 plane guidance with probe tracing. I would rather add this as an AI assist feature rather than adding it as a core feature since OMEA already has a checklist for anatomy as displayed in the image on its website.

INPUTS

1. Live ultrasound B-mode feed (DICOM stream)

2. Probe position/orientation

3. Patient metadata:

   - GA (14-24 weeks)

- Maternal BMI

- Exam type (routine anatomy)

4.Button click to activate this AI assistant feature

5. Current anatomy detection (OMEA's existing AI)

TRIGGERS:

1.Click activation and annotation on screen will lead to instant anatomy measurement

2. Another button to turn on ISUOG guidance protocol to trace probe accordingly

OUTPUTS:

1.) On screen auto measurements captured

2.) Probe tracing path shown on the screen based on the probe position for following ISUOG's standard

HUMAN-IN-LOOP-POINTS:

1.) Complete control for the sonographer to activate any of the two features with a button click.

FAILURE CASES AND FALLBACKS

Scenario	Detection	Fallback
Anatomy measurement incorrect	Sonographer feels that the measurement is wrong	Retake the measurement or switch back to manual mode
Path tracing lost	Probe leaves skin	Blinks on screen to show "Reapply probe to Resume"

What happens when the system is wrong?

In case the measurements are wrong or probe tracing is wrong the scan has to be redone again which will increase the time to finish the process. This is a huge drawback but with enough training should have a very minimal fail rate.

Topic 9: Expansion Strategy

Deliverable K: Expansion Market Selection

1.)Which practice type or geographical market would you prioritise for expansion following the initial launch and what factors make this expansion strategically sound?

- According to this resource, the Australian market seems to have more potential buyers in the near future.
- The Australia ultrasound devices market also gains from a supportive regulatory environment after the Therapeutic Goods Administration (TGA) streamlined audits for low-risk equipment in 2024, easing time-to-market for global manufacturers.
- Two of the eye catching trends in the resource mentioned above was Rapid Adoption of POCUS in Primary Care & Ambulances, Miniaturization & AI-Assisted Imaging Workflow

2.)What new constraints or considerations would this expansion introduce?

- Approval from TGA would still be needed which is similar to FDA approval
- They have something called the Medicare Benefits Schedule (MBS) which is an Australian Government-managed list of medical services—such as consultations, tests, and procedures—that are subsidized, determining the rebate patients receive but medical software must utilize the Medicare Benefits Schedule by embedding MBS item logic into clinical workflows, ensuring services meet billing rules, guiding compliant documentation, and enabling accurate, auditable claims submission.

Topic 10: Reflection

Deliverable L: Reflection

1.)Which assumption did you find most fragile or difficult to justify, and why?

The hardest part for me to defend is that the overall scan time is the source of friction. Scan time is highly variable based on things that are largely outside the control of the sonographer—fetal position, maternal BMI, fetal activity—so it's not a fixed measure of friction by itself. It sometimes even depends on the sonographer's experience level.

Looking deeper, the friction isn't just that scan time is longer, but that the process of scanning is complex and that you have to reimage so frequently. This study illustrates that the process of

scanning for anomalies doesn’t follow a clean, predictable pattern. Instead, sonographers find themselves going back over the same areas of the body multiple times to get images that are good enough to make a diagnosis.

This means that even experienced sonographers have to go back over parts of the scan because of clinical limitations, not because they’re doing something wrong. The friction is both physical (reimaging, longer scan times) and cognitive (orienting yourself and making sure that nothing is missed in a non-linear process). The challenge isn’t to reduce scan times but to reduce unnecessary rework while maintaining diagnostic accuracy and image quality.

2.)What aspects of the problem were most challenging to reason about without direct access to clinicians?

To be blunt, the scarcity of the data for private practices was the most challenging for me. I believe that is a niche and hence makes sense. So most of the data that i have written are mostly educated guesses and some blogs and resources that helped me understand the picture altogether.

3.)If you were to join the team, what would you prioritise validating first, and for what reasons?

I would first try to see how real business is done. I would first see the product’s features and see the friction points that you are prioritizing and see if it's in line with what I have prioritized. I would see which practice type was chosen and targeted which is bringing revenue to origin.